Credit Card Fraud Detection Using Machine Learning

Executive Summary

This project focuses on developing a machine learning solution capable of identifying fraudulent credit card transactions within a highly imbalanced financial dataset. The dataset contains over 1.29 million simulated credit card transactions, where .fraudulent activities account for less than 1% of all records

Instead of relying solely on raw transaction attributes, the project emphasizes feature engineering by creating more than 10 domain-driven features derived from customers' historical behavior, transaction timing, and geographical information. These engineered features significantly improve the model's ability to distinguish .fraudulent transactions from legitimate ones

:The final model, built using XGBoost, achieved

Precision: 95% •
Recall: 80% •
F1-Score: 87% •

These results demonstrate a strong balance between maximizing fraud detection and minimizing false alarms, making the solution practical for real-world financial .systems where both fraud prevention and customer experience are critical

Business Problem

Financial fraud remains one of the most significant challenges faced by banks and payment service providers. Fraudulent transactions represent only a tiny fraction of all financial transactions, making fraud detection a classic highly imbalanced .classification problem

Traditional rule-based systems often fail to capture complex fraud patterns, while manual investigation is impractical due to the enormous transaction volume. Consequently, machine learning provides a scalable and intelligent approach .capable of learning hidden behavioral patterns from historical data

The primary objective of this project is to develop a predictive model that accurately identifies fraudulent transactions while minimizing the number of legitimate

.transactions incorrectly flagged as fraud

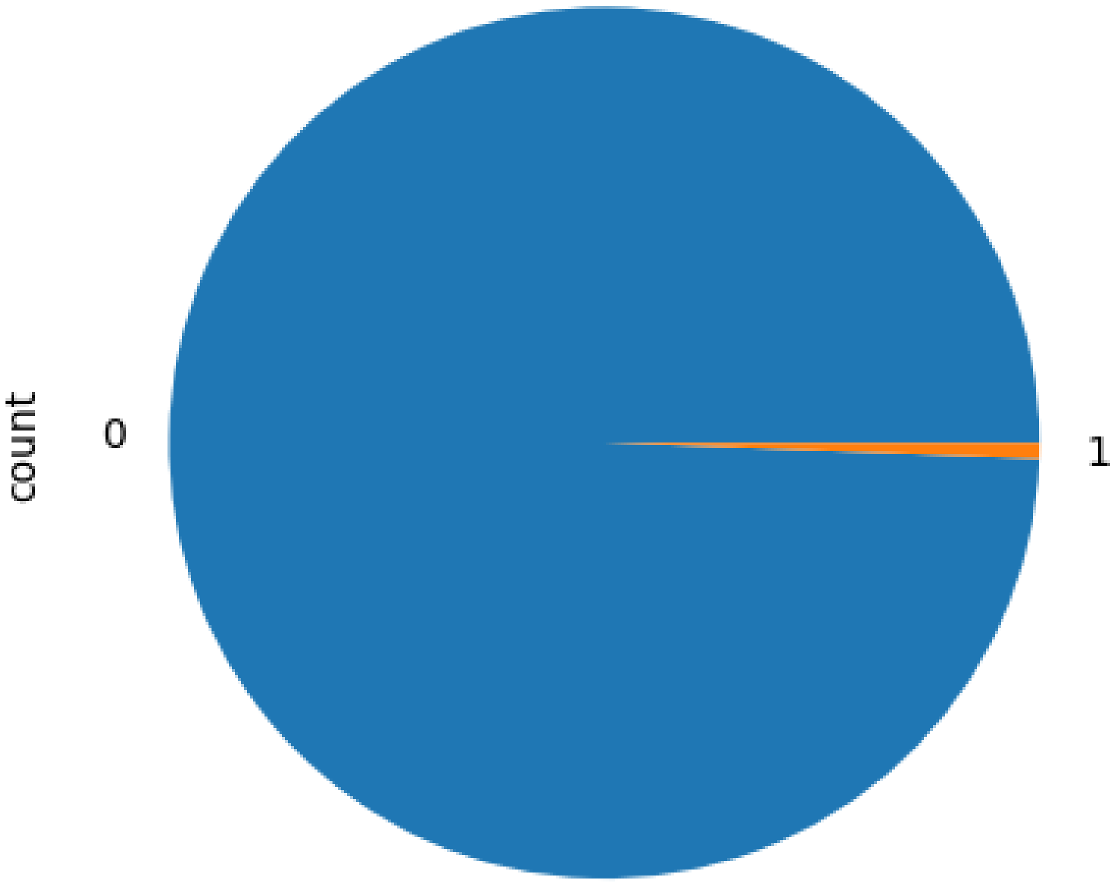
Dataset Overview

The project utilizes a simulated credit card transaction dataset with the following :characteristics

Attribute	Value
Total Transactions	1,296,675
Original Features	22
Data Types	Customer, Merchant, and Transaction Information
Fraud Rate	Less than 1%

The extreme imbalance between fraudulent and legitimate transactions makes conventional accuracy an unreliable evaluation metric. Therefore, the project focuses on metrics that better reflect fraud detection performance, including Precision, Recall, .and F1-Score

Figure 1 — Class Distribution



The class distribution illustrates the severe imbalance between legitimate and fraudulent transactions. Although fraud cases represent only a very small percentage of the dataset, they carry substantial financial risk, making effective detection both .challenging and essential

Exploratory Data Analysis (EDA)

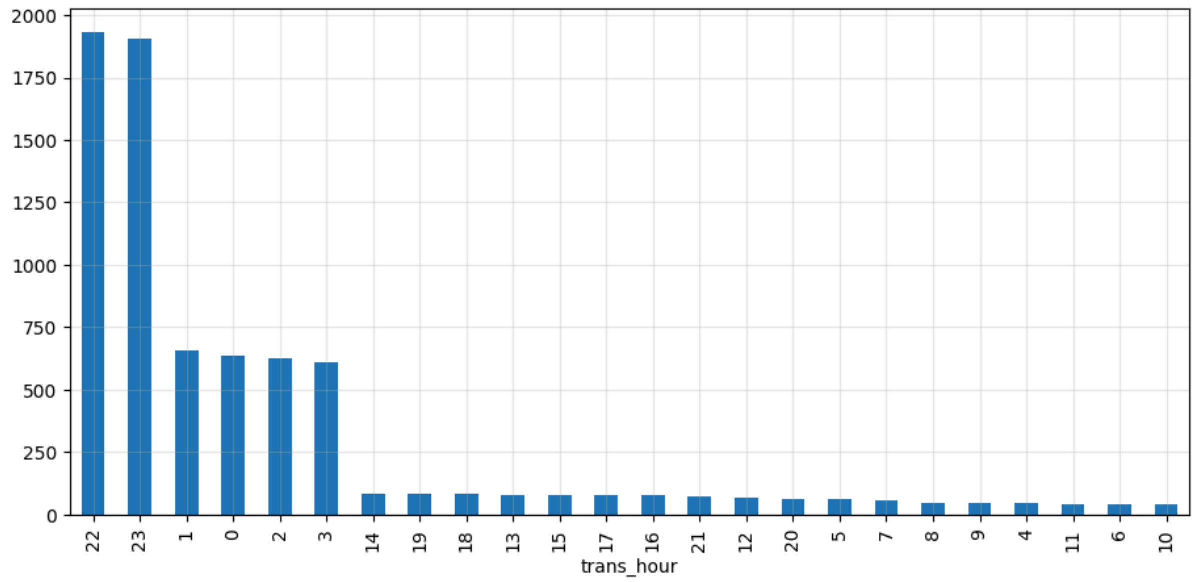
Before building the machine learning models, an extensive exploratory data analysis (EDA) was conducted to better understand when, where, and how fraudulent transactions occur. These insights guided the feature engineering process by identifying meaningful behavioral patterns rather than relying solely .on raw transaction attributes

Fraud Distribution by Time of Day

The hourly analysis revealed that fraudulent transactions are not uniformly distributed throughout the day. Instead, they are highly concentrated during **late-night and early-morning hours**, particularly between **10:00 PM and 3:00 .AM**

This pattern suggests that fraudsters tend to exploit periods when transaction .monitoring is less active and legitimate customer activity is relatively low

Figure 2 — Fraud Transactions by Hour of Day

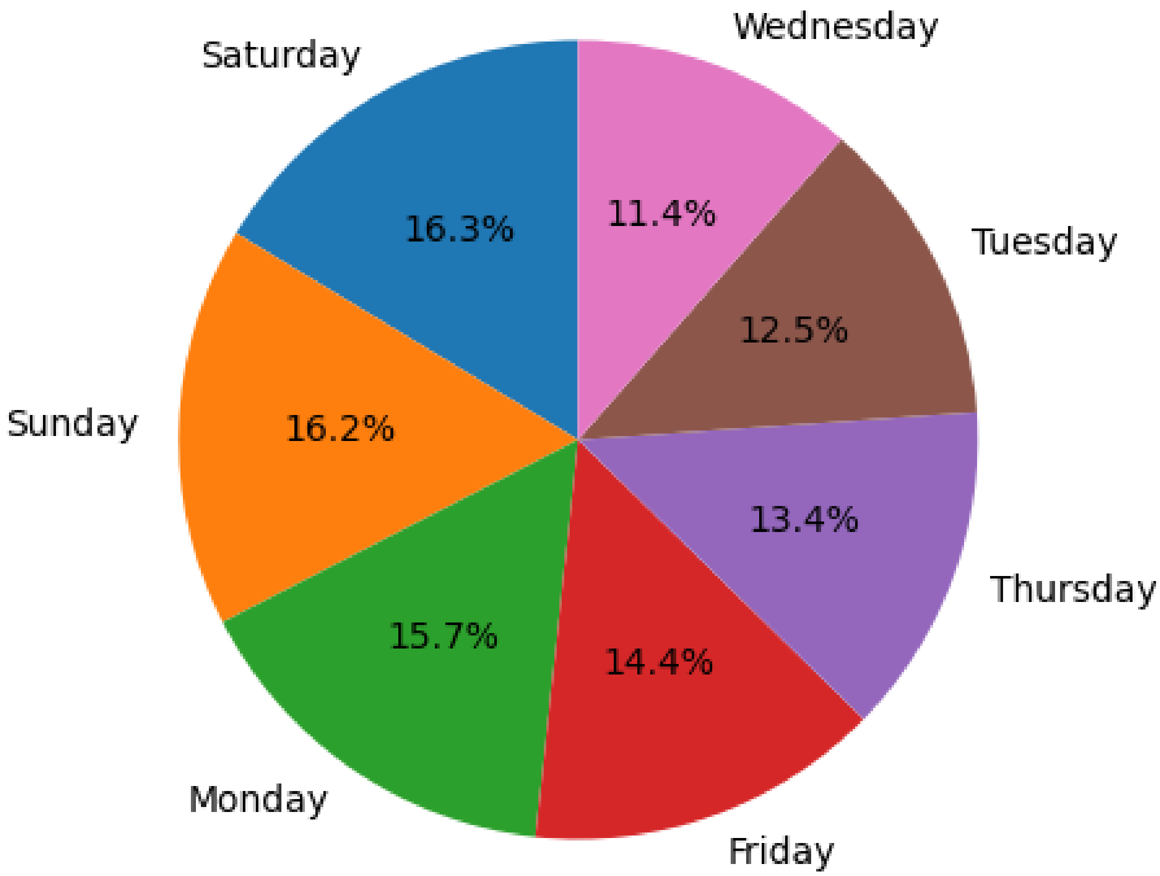


Fraud Distribution by Day of Week

Analyzing transaction activity across the week showed that fraud incidents .increase noticeably during weekends

Saturday and **Sunday** recorded the highest number of fraudulent transactions compared to weekdays, indicating that fraudulent behavior may be influenced by changes in customer spending habits and reduced operational monitoring .during weekends

Figure 3 — Fraud Transactions by Day of Week



Fraud Distribution by Month

.Monthly analysis highlighted seasonal variations in fraud activity

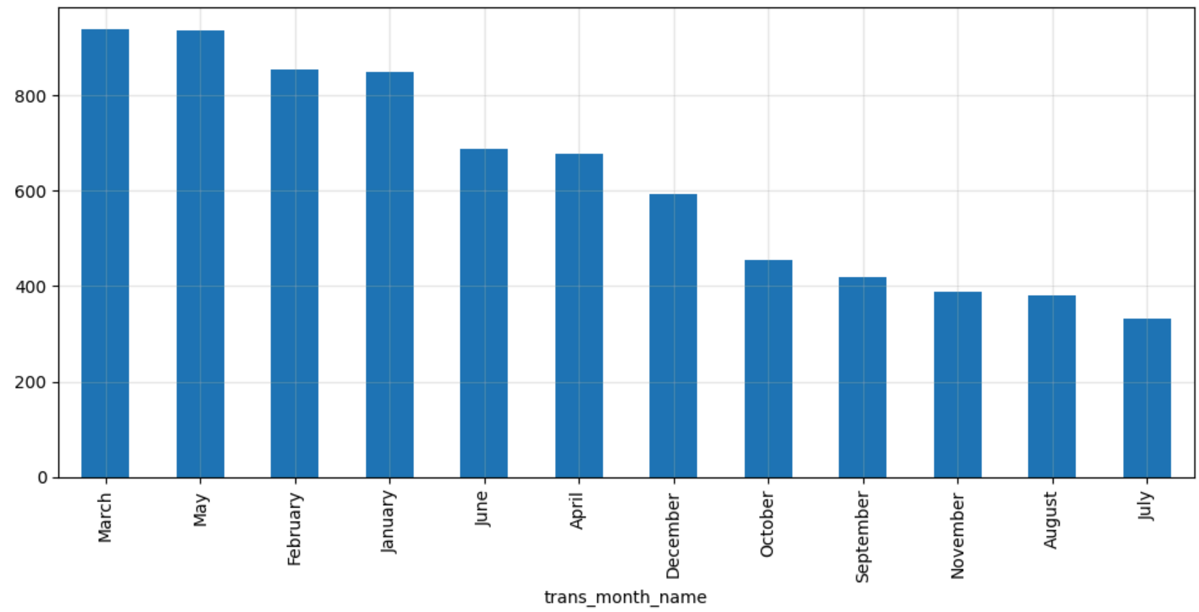
:The highest fraud frequencies were observed during

- January
- February
- March
- May

.In contrast, fraud activity declined during the summer months

These seasonal trends may reflect changes in consumer purchasing behavior, promotional periods, or other temporal factors that influence transaction .patterns

Figure 4 — Fraud Transactions by Month



Fraud by Merchant Category

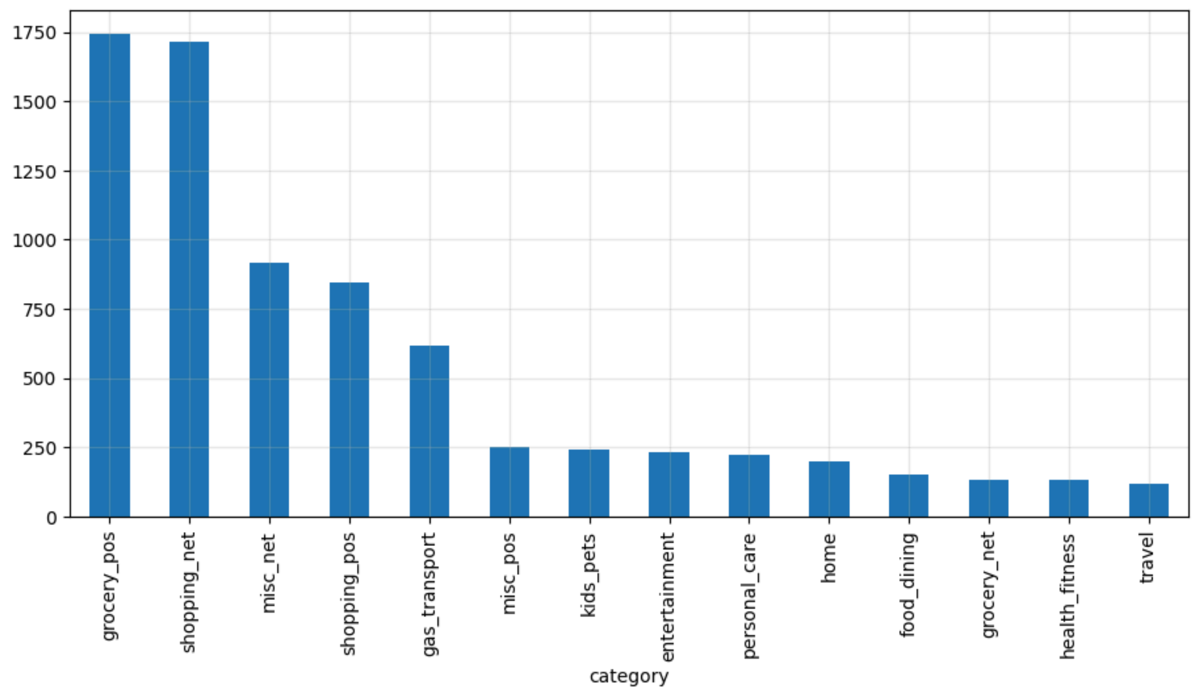
.Fraud occurrences vary significantly across merchant categories

The analysis showed that the following categories experienced the highest :number of fraudulent transactions

- Online Shopping (net_shopping)
- Grocery Point-of-Sale (pos_grocery)

This observation indicates that certain merchant categories are inherently more vulnerable to fraudulent activity and should receive additional attention during .model development

Figure 5 — Fraud Distribution Across Merchant Categories

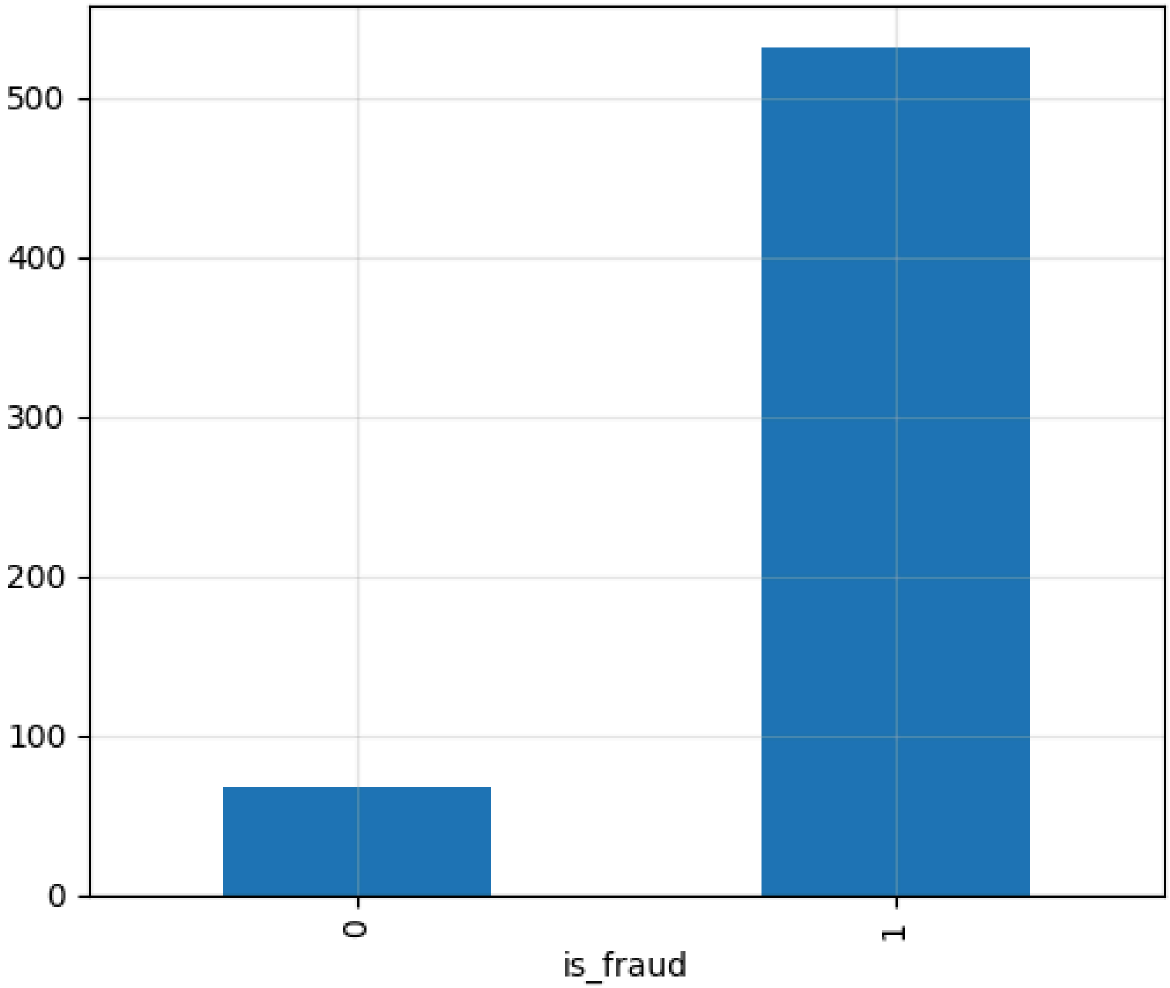


Transaction Amount Analysis

Comparing transaction amounts between legitimate and fraudulent .transactions revealed a clear distinction

On average, fraudulent transactions involve **higher monetary values** than legitimate ones. This finding suggests that transaction amount is an informative .predictor and should be incorporated into the fraud detection model

Figure 6 — Average Transaction Amount: Legitimate vs. Fraudulent



Feature Engineering

A key objective of this project was to move beyond raw transaction data by creating meaningful features that capture customer behavior, transaction .history, temporal patterns, and geographical information

Instead of allowing the model to infer these relationships implicitly, domain knowledge was used to engineer features that directly represent suspicious .behaviors commonly associated with financial fraud

:The following engineered features were developed

Engineered Feature	Source	Purpose

hour_trans	Transaction Timestamp	Capture fraud patterns associated with specific hours of the .day
trans_weekday	Transaction Timestamp	Identify variations in fraud activity across .different weekdays
trans_month	Transaction Timestamp	Capture seasonal .fraud trends
age_customer	Date of Birth	Measure customer age and identify demographic risk .patterns
merchant_customer_distance	Customer & Merchant Coordinates	Measure the geographical distance between the customer .and the merchant
txn_last_distance_max	Location + Transaction History	Detect impossible travel scenarios between consecutive .transactions
h1_count_txn_customer	Customer History	Count the number of customer transactions within the previous hour to identify unusually frequent .activity
customer_avg_amt_so_far	Customer Transaction History	Compute the customer's historical average transaction amount using only .past transactions

amt_deviation_from_avg	Current Transaction + Customer History	Measure how much the current transaction deviates from the customer's normal .spending behavior

Preventing Data Leakage

Special attention was given to preventing **data leakage** during feature .engineering

All historical features were calculated using **only transactions that occurred before the current transaction**. Future information was never used, ensuring that the model accurately simulates a real-world fraud detection environment .and produces reliable evaluation results

Model Development

To establish a performance baseline, a **Logistic Regression** model was initially trained using the engineered dataset. While the baseline model provided reasonable results, it struggled to capture the complex, nonlinear relationships .present in fraudulent transaction patterns

To improve predictive performance, **XGBoost** was selected as the primary :classification model due to its ability to

- .Handle highly imbalanced datasets effectively
- .Capture complex nonlinear relationships between features
- .Reduce overfitting through regularization
- .Deliver high predictive accuracy with efficient training

The model was trained using the engineered features and evaluated on an .unseen test dataset

Model Evaluation

,Because fraudulent transactions represent less than **1%** of the dataset traditional **Accuracy** is not an appropriate evaluation metric. A model could achieve over **99% accuracy** simply by predicting every transaction as legitimate .while failing to detect fraud entirely

Instead, the model was evaluated using metrics that better reflect fraud :detection performance

Metric	Score	Interpretation
Precision	**95%**	of transactions 95% flagged as fraud were .actually fraudulent
Recall	**80%**	The model successfully detected 80% of all fraudulent .transactions
F1-Score	**87%**	Demonstrates a strong balance between .precision and recall

These results indicate that the model is capable of detecting the majority of .fraud cases while generating a relatively small number of false alerts

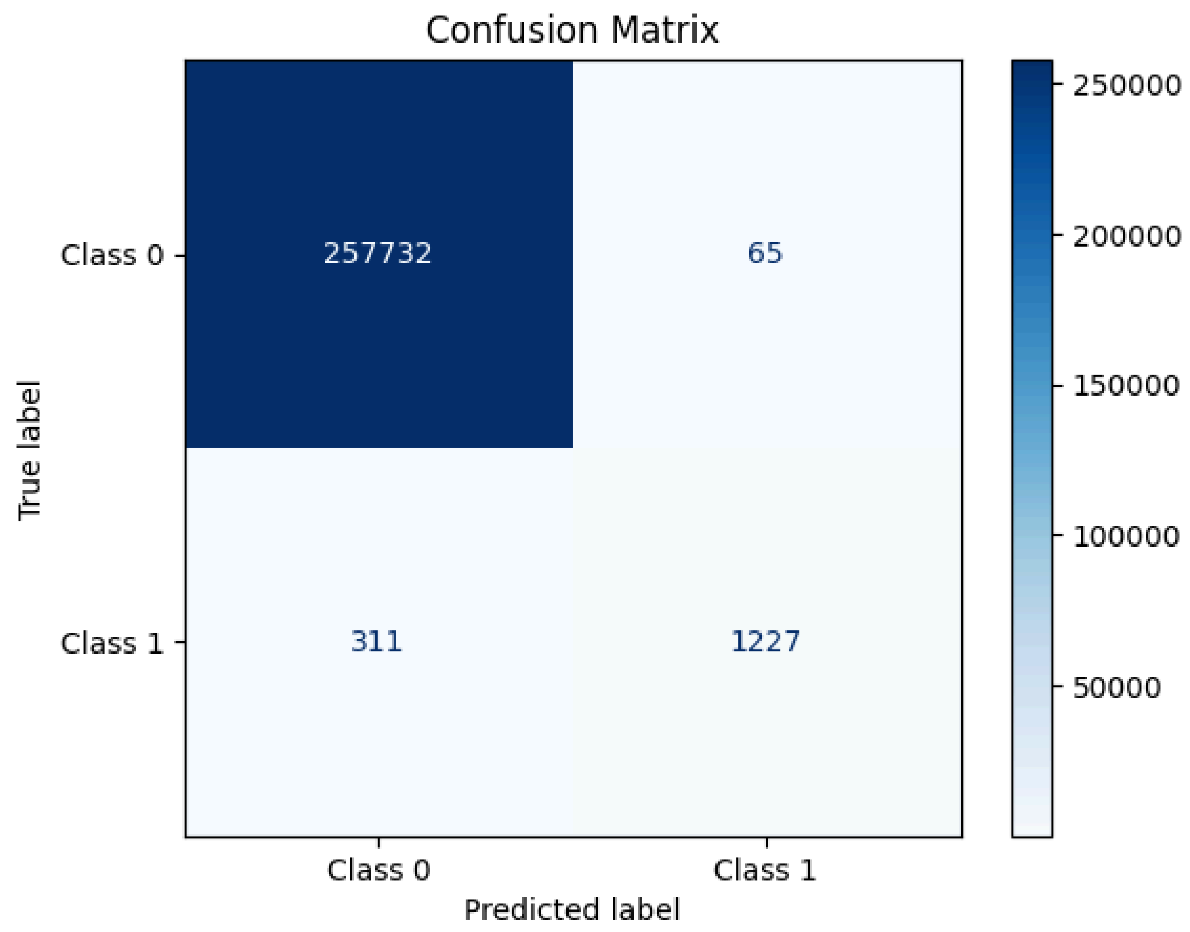
Confusion Matrix Analysis

The confusion matrix provides a more detailed understanding of the model's .performance

Out of **1,538 actual fraudulent transactions**, the model correctly identified .**1,227 fraud cases**, resulting in a **Recall of 80%**

In addition, only **65 legitimate transactions** were incorrectly classified as fraudulent out of more than **257,000 legitimate transactions**, demonstrating a .very low false-positive rate

Figure 7 — Confusion Matrix



Why Accuracy Was Not Used

Accuracy was intentionally excluded as the primary evaluation metric because it .is misleading for highly imbalanced classification problems

For example, a model that predicts every transaction as legitimate would still achieve extremely high accuracy due to the overwhelming number of non-.fraudulent transactions, despite failing to detect fraud altogether

Therefore, **Precision**, **Recall**, and **F1-Score** provide a much more realistic .assessment of model performance

Feature Importance

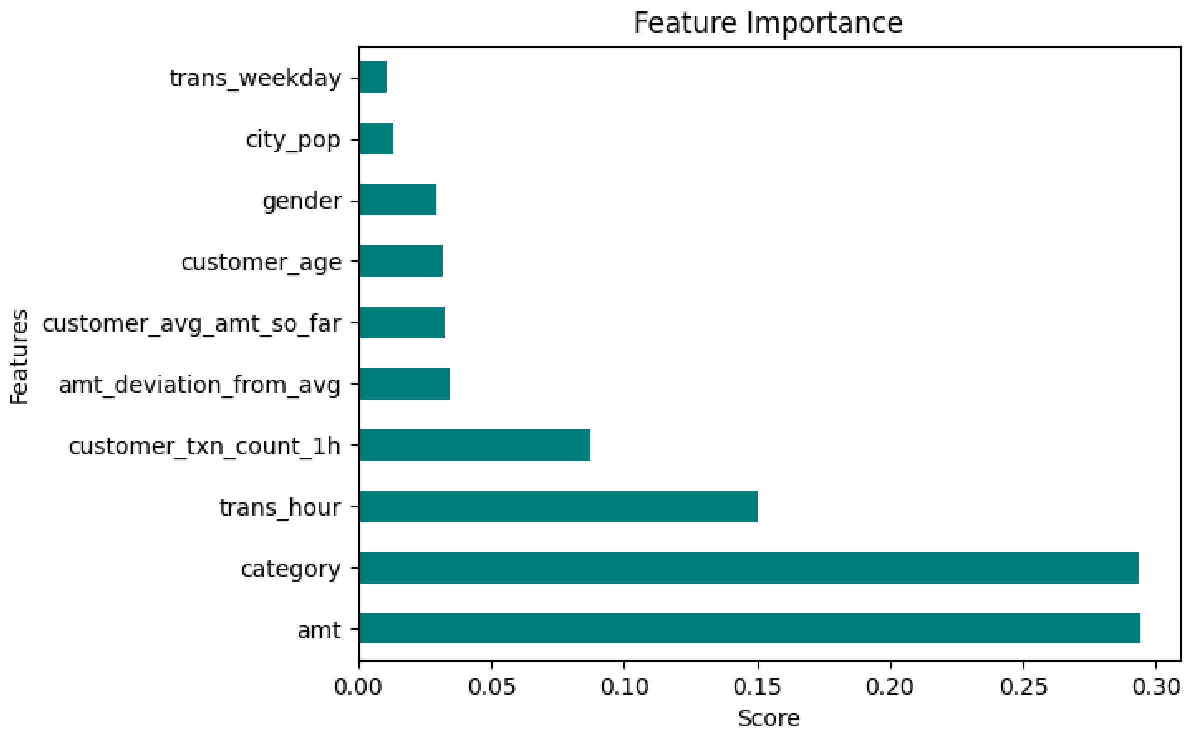
Feature importance analysis was performed to identify the variables that .contributed most to the model's predictions

:The three most influential features were

Transaction Amount (amt) .1
Merchant Category (category) .2
Transaction Hour (hour_trans) .3

These findings align with the exploratory data analysis, confirming that transaction value, merchant type, and transaction timing are among the .strongest indicators of fraudulent behavior

Figure 8 — Feature Importance



Business Perspective: Threshold Optimization

Choosing the classification threshold is not only a technical decision but also a .business decision

:Increasing the fraud detection sensitivity results in

.Detecting more fraudulent transactions •
.Increasing the number of false alarms •
.Potentially interrupting legitimate customer transactions •

:Conversely, decreasing the sensitivity leads to

- .Fewer false alarms
- .A smoother customer experience
- .Increased risk of undetected fraudulent transactions

.The optimal threshold depends on business priorities and risk tolerance

In its current configuration, the model achieves an effective balance by detecting **80% of fraudulent transactions** while generating only **65 false positives**, making it suitable as a practical starting point for deployment in real-.world fraud detection systems

Conclusion

This project demonstrates the complete lifecycle of building a production-.oriented fraud detection system using machine learning

Rather than relying solely on raw transactional data, the project emphasizes domain-driven feature engineering to capture customer behavior, temporal patterns, and geographical relationships. These engineered features .significantly enhanced the model's predictive capability

:The final **XGBoost** model achieved

- **Precision 95%**
- **Recall 80%**
- **F1-Score 87%**

The results indicate that the proposed solution successfully balances fraud detection performance with customer experience, making it well-suited for financial institutions seeking to minimize fraud losses while reducing .unnecessary transaction interruptions

Beyond predictive performance, the project highlights the importance of proper feature engineering, handling class imbalance, selecting meaningful evaluation .metrics, and aligning machine learning decisions with real business objectives

Prepared by: Machine Learning Project – Credit Card Fraud Detection